

KAAE UNIVERSITY COLLEGE

MID-SEMESTER EXAMS

COURSE CODE/TITLE: KNUR 110-MICROBIOLOGY, INFECTION PREVENTION &
CONTROL

Name,

Index no:

Program,

Answer all question

1. Which of the following is it characteristie of Gram-negative bacteria?

A) Thick peptidoglycan layer

B) Presence of teichoie acids

C) Lipopolysaccharide layer

D) High peptidoglycan content

2. Which of the following microorganisms is considered an obligate aerobe?

A) Escherichia coli

B) Clostridium botulinum

C) Pseudomonas aeruginosa

D) Helicobacter pylori

3. Which phase of the bueterial growth curve is characterized by rapid cell division?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death (Decline) Phase

4. What is the function of the bacterial cell wall in relation to osmotic pressure?

A) To facilitate binary fission

B) To provide energy

C) To prevent the cell from bursting

D) To store nutrients

5. In which phase of the bacterial growth curve does the number of dying cells exceed the number of new cells being formed?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death (Decline) Phase

6. Which of the following best describes bacterial with rod-like shape?

A) bacilli

B) cocci

C) spirilla or spirochetes

D) cubilli

7. What is the main component of the bacterial cell wall?

A) Lipopolysaccharides

B) Teichoic acids

C) Peptidoglycan

D) Phospholipids

8. Which type of bacteria has a thick peptidoglycan layer?

A) Gram-positive bacteria

B) Gram-negative bacteria

C) Mycoplasmas

D) Cyanobacteria

9. What substance in Gram-negative bacteria is responsible for its toxic effects?

A) Peptidoglycan

B) Teichoic acid

C) Lipid A

D) Periplasm

10. Which bacteria can thrive in temperatures between 15-20°C?

A) Thermophiles:

B) Psychrophiles

C) Mesophiles

D) Extreme thermophiles

11. Bacteria that require 21% O₂ for their growth are called

A) Microaerophilic

b) Obligate aerobes

C) Acidophiles

D) Alkaliphiles

12. Which bacteria thrive in acidic environments?

A) Alkaliphiles

B) Neutralophiles

C) Psychrophiles

D) Acidophiles

13. Which phase is characterized by no apparent cell division but high metabolic activity as bacteria adapt to a new environment?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death: (Decline) Phase

14. In which phase of the bacterial growth curve does the number of dying cells exceed the number of new cells being formed?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death: (Decline) Phase

15. During which phase might bacterial infection produce the most severe symptoms due to toxin production?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death: (Decline) Phase

16. At what phase are antibiotics generally most effective against bacteria?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death: (Decline) Phase

17. In which phase do bacteria begin to form endospores or other survival structure due to nutrient depletion?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

D) Death: (Decline) Phase

18. Which is the primary reason for the cell death during the death (Decline) Phase ?

A) Lack of metabolic activity

B) Rapid cell division

C) Nutrient depletion and waste accumulation

D) High metabolic activity:

19. Which phase of the bacterial growth curve involves the synthesis of enzymes and molecules necessary for bacterial growth?

A) Lag Phase

B) Log (Exponential) Phase

C) Stationary Phase

d) Death (Decline) Phase

20. The destruction of all forms of microbial life, including bacterial spores is termed

a) Sanitization

b) Disinfection

c) Antisepsis

d) Sterilization

21. The physical removal of microorganisms using soaps or detergents is termed

a) degerming

b) disinfection

c) sanitization

d)sterilization

22. What is u germicide?

a) A chemical agent that inhibits microbial growth

b) A chemical agent that kills microorganisms

c) A chemical agent that only kills fungi

d) A chemical agent that removes microorganisms physically

23, A chemical agent that prevents the multiplication of bacteria without necessarily killing them may refer to

a) Bactericide

b Fungicide

c) Bacteriostatic agent

d) Germicide

24. Why are germicides more effective at high temperatures?

a) Molecules have higher kinetic energy, increasing the reaction rate

b) Molecules are more stable at high temperatures

c) Microorganisms are less resistant at high temperatures

d) High temperatures inhibit microbial growth

25. In a mutualistic association

a) one organism benefits and the other is harmed

b) one organize benefits and the other is unaffected

c) both organisms benefit

d) both organisms are harmed

26, What is the difference between disinfection and antisepsis?

1) Disinfection kills all microorganisms, while antisepsis kills only bacteria.

b) Disinfection is used on inanimate objects, while antisepsis is used on living tissues.

5) Disinfection is used for sanitization, while antisepsis is used for sterilization.

d) Disinfection and antisepsis are the same process:

27. What is the term for the lowest temperature at which all microorganisms are killed in 10 minutes?

a) Thermal death point

b) Thermal death time

c) Decimal reduction time

d) Denaturation temperature

28. Which method is used to sterilize heat-sensitive materials as vaccines?

a) Autoclaving

B) Incineration

c) Filtration

d) Dry heat

29. The reduction of pathogens to a level deemed safe by public health guidelines refer to ?

a) Sterilization

b) Sanitization

c) degerming

d) antisepsis

30. At what temperature and pressure does an autoclave operate to achieve sterilization?

a) 100 °C at 15 psi

b) 121 °C at 15 psi

c) 150 °C at 20 psi

d) 200 °C at 25 psi

31. Which of the following is a double-stranded DNA virus?

a. SARS-CoV-2

b. MERS-CoV

c. Human papillomavirus

d. rhinovirus

32. Which of the following bacteria will give positive results for a coagulase test?

a. *S. epidermidis*

b. *S. saprophyticus*

c. *S. aureus*

d. *S. hominis*

33. Which enzyme is produced by *S. aureus* coagulates plasma resulting in fibrin deposition and interference with phagocytosis?

a. Lipase

b. Haemolysin

c. Coagulase

d. Hyaluronidase

34. Which of these organisms is a cestode?

a. *Fasciola hepatica*

b. Tape worm

c. *Paragonimus westermani*

d. Guinea worm

35. Which of the following is a sexually transmitted infection caused by a spirochete bacterium?

a. Chlamydia

b. Gonorrhea

c. Syphilis

d. Herpes

36. What is the function of ribosomes in a bacterial cell?

a. Synthesizing DNA

b. Producing ATP

c. Protein synthesis

d. Cell division

37. Which of the following culture methods is best for the agglutination test?

a. Stroke culture

b. Strek culture

c. Lawn culture

d. Stab culture

38. Which of the following is positive for the oxidase test?

a. Pseudomonas

b. Salmonella

c. Shigella

d. All of the above

39. Which of the following is indole positive?

a. Klebsiella

b. Salmonella

c. E. coli

d. All of the above

40. Which of the following is positive for the urease test?

a. Proteus

b. Salmonella

c. Klebsiella

d. a and c

41. which of the following is/are positive for the catalase test?

a. Listeria

b. Pseudomonas

c. C. diphtheria

d. All of the above

42. Which of the following is a lactose fermenter?

a. E. coli

b. Salmonella

c. Shigella

d. Proteus

43. The capsid and the nucleic acid in the viral structure are referred to as

a. Envelope

b. Nucleocapsid

c. Nucleoside

d. nucleosome

44. Which of the following is an enveloped virus?

a. SARS-CoV-2

b. Human Papillomavirus

c. Poliovirus

d. All of the above

45. Which of the following is a naked virus?

a. HIV

b. Ebola

c. SARS-CoV-2

d. Human Papillomavirus

46. The SARS-CoV-2 attaches itself to the host through which protein?

a. S glycoprotein

b. Envelope

c. Nucleocapsid

d. Membrane

47. Which of the following is a viral vector vaccine?

a. AstraZeneca

b. Sinopharm

c. Sinovac

d. Pfizer

48. The Varicella Zoster Virus (VZV) is an example of

a. Rotavirus

b. Herpesvirus

c. Parvovirus

d. All of the above

49. The individual units of the viral capsid are known as

a. Capsid protein

b. Capsomers

c. Tegument

d. Envelope

50. Which of the following fungi is said to be opportunistic?

a. Aspergillus

b. Candida

c. Cryptococcus

d. All of the above

51. Which of the following is a true fungal pathogen?

a. Aspergillus

b. Coccidioides

c. Cryptococcus

d. all of the above

52. Histoplasmosis is a condition caused by

a. Histoplasma capsulatum

b. Histoplasma neoformans

c. Cryptococcus

d. Aspergillus

53. Coccidioidomycosis is caused by

a. C. immitis

b. C. immanitis

c. Cryptococcus

54. Histoplasma What is the primary method of transmission for Salmonella infections?

a. Vector-borne transmission

b. Fecal-oral transmission

c. Droplet transmission

d. Sexual contact

55. All the following Treponema species cause diseases in humans except:

a. T. carateum

b. T. pallidum

c. T. pertenuis

56. T. paraluisuniculi Which microorganism(s) perform photosynthesis by utilizing light?

a. Cyanobacteria, fungi, and viruses

b. Viruses

c. Cyanobacteria

d. Fungi

57. Which of the following is a Gram-positive bacterium?

a. Staphylococcus

b. Streptococcus

c. Neisseria

d. Bacillus

58. What is the primary function of capsules in bacterial cells?

a. Facilitate cell division

b. Aid in respiration

c. Protect against antibiotics

d. Aid attachment to host cells and protect against host defenses

59. Which of the following is a bacterial toxin that causes muscle paralysis by blocking acetylcholine release at neuromuscular junctions?

a. Botulinum toxin

b. Tetanus toxin

- c. Cholera toxin
- d. Diphtheria toxin

60. Which hosts are involved in the life cycle of Plasmodium parasites?

- a. Mosquitoes only

b. Humans and mosquitoes

- c. Birds and mosquitoes

61. Insect and mosquitoes Which family do herpes viruses belong to?

a. Herpesviridae

- b. Flaviviridae
- c. Adenoviridae
- d. Papillomaviridae

62. What term describes the process where blood leukocytes engulf disease-causing bacteria?

a. Phagocytosis

- b. Vaccination
- c. Attenuation
- d. Pinocytosis

63. All the following are gram-positive spherical bacteria, except

a. S. aureus

- b. S. epidermidis
- c. E. faecalis
- d. E. coli.

64. Bacteria with flagella clusters at both cell poles are known as?

- a. Amphitrichous
- b. Monotrichous
- c. Lophotrichous

d. Peritrichous

65. Which among the following is a spirochete?

a. Treponema pallidum

b. *Streptomyces* sp.

c. *Streptococcus viridans*

d. *Corynebacterium diphtheriae*

66. Common infections of *Cryptococcus neoformans* occurs in

a. AIDS patients

b. Cancer patients

c. Diabetic patients

d. All of the above

67. Filariasis is a condition caused by

a. *Wuchereria bancrofti*

b. *Ascaris lumbricoides*

c. *Brugia malayi*

d. *Ancylostoma duodenale*

68. Which of the following has a capsule around the cell wall?

a. *Klebsiella pneumonia*

b. *Vibrio cholerae*

c. *Neisseria*

d. All of the above

69. Which of the following is a psychrophile?

a. *Pseudomonas fluorescens*

b. *E. coli*

c. *Bacillus stearothermophilus*

d. *Staphylococcus aureus*

70. Which of the following is a mesophile?

a. *E. coli*

b. *Pseudomonas fluorescens*

c. *Thermococcus*

d. *B. stearothermophilus*

